

Scale or Rotation? Decomposing Neural Tangent Kernel Movement During Grokking

Group G2

Zain Al-Saffi (s4800836) · Jasper Chong (s4883420) · Aaron D’Mello (s4890257)

Zac Kienzle (s4803349) · Samuel Rieger (s4904959)

[TODO: Need to verify all citations by hand]

1 Topic

The task is modular addition. The input concatenates the one-hot codes of $a, b \in \{0, \dots, p-1\}$ and the target is the one-hot code of $(a + b) \bmod p$. A network $f(x; \theta)$ trained on a fraction of the p^2 pairs *groks* when its training loss falls early but its test loss stays at chance for orders of magnitude longer before falling abruptly (Power et al., 2022). The grokking time t_{grok} counts the training steps from when the training loss first falls below a fixed level τ_{train} to when the test loss first falls below a fixed level τ_{test} . The neural tangent kernel (NTK) of f is $\Theta_t(x, x') = \langle \nabla_{\theta} f(x; \theta_t), \nabla_{\theta} f(x'; \theta_t) \rangle$, and K_t is its empirical version on a fixed probe set of n inputs. Training is *lazy* when Θ_t stays at Θ_0 , so the network behaves as its linearisation, and *rich* when Θ_t moves (Chizat, Oyallon, and Bach, 2019; Jacot, Gabriel, and Hongler, 2018). The leading account of grokking is a lazy-to-rich transition: a linearised model memorises, but generalising needs the kernel to move (Kumar et al., 2024).

Saying that the kernel moves bundles two claims. A kernel can change in overall *scale*, with every eigenvalue rescaled and the eigenbasis fixed, or it can *rotate*, so that the eigenbasis itself changes. Let $\hat{K} = K/\|K\|_F$ be the normalised kernel, and define the scale term $S_t = \log(\|K_t\|_F/\|K_0\|_F)$ and the rotation term $R_t = 1 - \langle \hat{K}_t, \hat{K}_0 \rangle_F$. The standard measure of kernel change mixes the two, since it decomposes as

$$\|K_t - K_0\|_F^2 / \|K_0\|_F^2 = e^{2S_t} + 1 - 2e^{S_t}(1 - R_t). \quad (1)$$

The distinction matters because the theorem usually cited to link weight decay to kernel movement describes pure scaling. A network is k -homogeneous if scaling every parameter by c scales its output by c^k ; a bias-free ReLU MLP has k equal to its depth. Lewkowycz and Gur-Ari (2020, Thms. 1–2) show that an infinitely wide k -homogeneous network trained with MSE and L_2 strength λ has kernel $\Theta_t = e^{-2(k-1)\lambda t} \Theta_0$. In our terms $S_t = -2(k-1)\lambda t$ and $R_t \equiv 0$: the kernel shrinks but never turns, so no features are learned. Yet Kumar et al. (2024, App. 13, arXiv v3) invoke this theorem to argue that weight decay ends lazy training, and hence that $t_{\text{grok}} \propto 1/\lambda$, an argument they call speculative and do not test on the kernel. Rotation is also necessary: a model that stays in the kernel regime provably needs a constant fraction of the p^2 pairs (Mohamadi et al., 2024), so at low training fractions the kernel must turn.

Research question. What sets t_{grok} ? The candidates are kernel rotation, the scale of the kernel or the weights, and the weight-decay rate. Three 2026 accounts each back a different one (§2), and none measures rotation. We measure all three on one grid and ask which t_{grok} tracks.

Our hypothesis H is that rotation is the clock. Let Y be the probe set’s one-hot label matrix and $H = I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top$ the centring matrix. The cosine A_t between HK_tH and $HYY^\top H$ is the centred alignment of Cortes, Mohri, and Rostamizadeh (2012). Rescaling K_t leaves it unchanged, so only rotation moves it. H says generalisation begins when A_t crosses a threshold A_* , that A_* is shared across α , $\eta\lambda$, N and p , and that t_{grok} is independent of S_t given A_t .

The prediction that separates H from its rivals comes from the exact kernel evolution $\dot{\Theta}_t = -2(k-1)\lambda \Theta_t + T_t + T_t^\top$ (Lewkowycz and Gur-Ari, 2020, Eq. S5, arXiv v1). The weight-decay term only rescales Θ_t , so A_t moves only through T_t . As T_t vanishes at infinite width (Jacot, Gabriel, and Hongler, 2018; Lewkowycz and Gur-Ari, 2020), rotation is a finite-width effect and a wider network turns more slowly. Under H the target A_* is fixed, so t_{grok} is a distance over

a rate: it should grow with N and stay flat in $\eta\lambda$. We fit $\log t_{\text{grok}} = c + a \log N + b \log(\eta\lambda)$ on a crossed grid. Under H, $a > 0$ and $b \approx 0$; under a weight-decay clock, $a \approx 0$ and $b \approx -1$; two nonzero exponents mean two clocks, the slower one setting the time.

2 Significance

Three 2026 accounts name three different clocks. Truong et al. (2026) intervene on the weight norm. Networks grok when the norm reaches a critical value $\|W\|_c \propto p^{0.38}$, and holding the norm at $\rho\|W\|_c$ makes t_{grok} grow exponentially in ρ , at a rate of about 7.5 for an MLP and 15 for unnormalised attention. Truong (2026) then shows that under cross-entropy this effect runs through the logit scale rather than the norm itself, but that this channel is absent under MSE, where about half of the effect is unexplained. Kim (2026) instead makes weight decay the clock, with $t_{\text{grok}} \propto (1 - \beta)/(\eta\lambda)$. All three are scale-like or optimiser-level quantities, so none can tell a rotating kernel from a decaying one; H supplies the missing variable. Our baseline uses MSE and vanilla gradient descent, where the logit-scale channel is absent, the theorem above applies as stated, and coupled L_2 regularisation coincides with decoupled weight decay.

The NTK–grokking link, the laziness knob α , eNTK spectral tracking, the norm delay law and norm mediation are all published. What is new here is decomposition (1) used as a timing variable, the (a, b) test that separates width-driven rotation from the λ -clock, and a rotation account of the norm law under MSE.

3 Feasibility

The baseline follows Kumar et al. (2024, App. 8.3, arXiv v3) exactly, so Tier 0 can be checked against a published figure. It uses one hidden layer of width $N = 100$, MSE loss, full-batch gradient descent at $\eta = 100$, $p = 23$, and 90% of the p^2 pairs for training. One setting of the swept hyperparameters is a cell, and the baseline is one cell. A cross-entropy arm at the baseline cell links our constants to Truong et al. (2026), whose results use cross-entropy.

Each checkpoint records the training and test MSE, the terms S_t , R_t and A_t on the centred kernel, the parameter movement, and $y^\top K^{-1}y$, which predicts when linearisation breaks. The empirical NTK on $n = 256$ probe points, summed over logits, takes seconds.

Three choices are fixed in advance. First, t_{grok} and its thresholds are pre-registered and sensitivity to τ is reported. A run has not grokked if its test loss is above τ_{test} when training stops. The run is kept, and t_{grok} is recorded only as longer than the run, which is right censoring. Second, laziness is set by the centred, rescaled predictor $\tilde{f} = \alpha[f(x, \theta) - f(x, \theta_0)]$ with $\eta = \eta_0/\alpha^2$ (Chizat, Oyallon, and Bach, 2019); since α , initialisation scale and label rescaling are one knob, we sweep only α . Third, weight decay is swept in $\eta\lambda$ below the stability bound $\eta\lambda < 2$.

The design has four arms. One is a 6×6 grid in $\alpha \times \eta\lambda$ at the baseline width. One is an $N \times \eta\lambda$ factorial with $N \in \{50, 100, 400, 1600\}$ and $\eta\lambda \in \{0, 10^{-3}, 10^{-2}, 10^{-1}\}$ for the (a, b) fit. One is a clamp arm with $\rho \in \{1, 1.25, 1.5, 1.75, 2\}$, and one is a generality arm with $p \in \{23, 29, 31\}$ and training fraction in $\{50, 70, 90\}\%$. Five seeds per cell, and ten on the clamp arm where the MSE effect is half-size, give about 350 runs and roughly 20 GPU-hours. **[TODO: time one run in week 1 and resize; does $p = 23$ grok cleanly? Colab/Kaggle or UQ Bunya?]**

The work falls into four tiers, each reportable on its own. Tier 0 reproduces the published α sweep and is a hard gate in week 1, since later work depends on it. Tier 1 measures S_t , R_t , A_t and t_{grok} over the grid, tests whether A_* is invariant, and exhibits a cell where kernel movement is large but A_t is flat. Tier 2, the head claim, is the (a, b) fit with bootstrap intervals and the residuals where a power law fails. Tier 3, a stretch goal, replicates the clamped-norm law under MSE and tests whether A_* is invariant to ρ . **[TODO: placeholder split]** Zac Kienzle builds the infrastructure, Aaron D’Mello runs Tier 0, Jasper Chong writes the kernel metrics, Samuel Rieger runs the sweeps and fits, and Zain Al-Saffi does the theory and write-up.

References

- Chizat, L., E. Oyallon, and F. Bach (2019). “On lazy training in differentiable programming”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv: 1812.07956.
- Cortes, C., M. Mohri, and A. Rostamizadeh (2012). “Algorithms for learning kernels based on centered alignment”. In: *Journal of Machine Learning Research* 13.28, pp. 795–828.
- Gromov, A. (2023). “Grokking modular arithmetic”. In: *arXiv preprint*. arXiv: 2301.02679.
- Jacot, A., F. Gabriel, and C. Hongler (2018). “Neural tangent kernel: Convergence and generalization in neural networks”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv: 1806.07572.
- Kim, T. (2026). “Grokking on the weight-decay clock: A rate hierarchy from softly broken symmetries”. In: *arXiv preprint*. arXiv: 2607.23967.
- Kumar, T. et al. (2024). “Grokking as the transition from lazy to rich training dynamics”. In: *International Conference on Learning Representations (ICLR)*. arXiv: 2310.06110.
- Lewkowycz, A. and G. Gur-Ari (2020). “On the training dynamics of deep networks with L_2 regularization”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv: 2006.08643.
- Mohamadi, M. A. et al. (2024). “Why do you grok? A theoretical analysis of grokking modular addition”. In: *International Conference on Machine Learning (ICML), PMLR 235*. arXiv: 2407.12332.
- Nanda, N. et al. (2023). “Progress measures for grokking via mechanistic interpretability”. In: *International Conference on Learning Representations (ICLR)*. arXiv: 2301.05217.
- Power, A. et al. (2022). “Grokking: Generalization beyond overfitting on small algorithmic datasets”. In: *ICLR MATH-AI Workshop*. arXiv: 2201.02177.
- Truong, X. K. (2026). “What does the weight norm control in grokking? Logit-scale mediation under cross-entropy”. In: *arXiv preprint*. arXiv: 2606.18465.
- Truong, X. K. et al. (2026). “The weight norm sets the grokking timescale: A causal delay law”. In: *arXiv preprint*. arXiv: 2606.13753.

Draft notes — delete this page before submission

Body is two pages with about one line to spare under Tectonic 0.17; references are excluded, per the brief, and now start on their own page. All citations were checked against primary sources on 31 Aug 2026. Under Tectonic the 31 Aug text ran ten lines onto a third page, so the 4 Sep rewrite was trimmed to fit. The bib entries were rebuilt in `ref.bib` and rechecked against the arXiv abstract pages on 4 Sep 2026. In a later 4 Sep pass the run-in bold headings were removed from every paragraph except the research question, the research question was rewritten in full sentences, and a few phrases were shortened so the body still fits. On 6 Sep the grokking time was spelled out as a count of steps between two threshold crossings, and “cell”, “has not grokked” and “right censoring” were each defined in plain words at first use. To pay for those lines the research question was tightened (“the grokking time” dropped after t_{grok} , “itself” dropped, “rotation of the kernel” became “kernel rotation”), the two compute TODOs were merged into one, “gradient-descent stability bound” became “stability bound”, and “not dropped” was folded into “the run is kept”. Page 2 is now full with no line to spare.

1. **What changed in this revision.** The hypothesis is now a *threshold* claim (A_* invariant, $t_{\text{grok}} \perp S_t \mid A_t$) plus a *rate-law* claim tested by estimating (a, b) in $\log t_{\text{grok}} = c + a \log N + b \log(\eta\lambda)$. That regression, not the old mediation test, is the head claim: it is the one experiment that separates the rotation clock, the scale clock and Kim’s weight-decay clock, and it fails informatively (both exponents nonzero $\Rightarrow$ two clocks). Tier 3 was demoted to a stretch replication because Truong (2026) already ran a norm mediation test under cross-entropy.
2. **Verified numbers** in Truong et al. (2026): $\|W\|_c = \{42, 48, 55, 61, 67\}$ for $p = \{29, 41, 59, 79, 97\}$, $\|W\|_c \propto p^{0.38}$; $t_{\text{grok}} \propto e^{\alpha p}$ with $\alpha \approx 7.4\text{--}7.5$ (MLP) against ≈ 15 (unnormalised attention), $R^2 > 0.99$; “the form of the delay law is cross-architectural while its rate is not”; LayerNorm removes the dependence. Their loss is cross-entropy — hence our CE arm. Truong (2026): logit-scale mediation $R^2 = 0.97$ under CE, channel absent under MSE, norm effect $\approx$ half size.
3. **Verified as stated:** Lewkowycz & Gur-Ari Thm. 1 and Thm. 2 (eigenvalues decay, eigenvectors static; assumes MSE, $k \geq 2$, gradient flow, infinite width, Dyer–Gur-Ari conjecture), Eq. (S5) with T_t from (S6) in arXiv v1 (these are (S8) and (S9) in v2, so the body cites the version); Kumar App. 8.3 config and App. 13 for weight decay (an earlier note said 14, but arXiv v3 and the ICLR camera-ready both number it 13, and App. 14 is momentum); Mohamadi et al. (ICML 2024, PMLR 235); Atanasov et al.; Gromov’s quadratic-activation closed form; Cortes et al., JMLR 13, 795–828. **Venue lines rechecked 4 Sep 2026:** Jacot (NeurIPS 2018), Chizat (NeurIPS 2019) and Lewkowycz (NeurIPS 2020) appear in the NeurIPS proceedings index; Nanda appears on OpenReview as ICLR 2023 (forum 9XFSbDPmdW); Power appears on the ICLR 2021 MATH-AI workshop page, while the arXiv posting is dated 2022. Kumar’s ICLR 2024 venue rests on the camera-ready checked on 31 Aug.
4. **Open decisions.** Coordinator sign-off (scope email to Dr Nan Ye before 14 Sep); AI-use statement here or in the report; whether “pre-registered” means public (OSF) or fixed in the repo before sweeps run; whether the CE arm is one cell or a full $\alpha \times \eta\lambda$ replication; whether to keep the p and training-fraction arms at all, given they buy generality rather than discrimination.
5. **Held for the 12-page report:** the derivation of (1); the tier table; uncentred-alignment and first-logit eNTK robustness variants; the censoring model; $y^\top K^{-1} y = \|w - w_0\|^2$ near the kernel limit; the confound demonstration figure. An abstract would cost about six lines here.
6. **Dropped from the body for space on 4 Sep 2026** (restore if lines are freed): “and cannot tell them apart”; “and so leaves A_t unchanged; alignment moves only through T_t ” (now “so A_t moves only through T_t ”); “and the usual summary statistic does not separate the two”; “each observes only a magnitude, which rescaling moves but rotation does not” (the point is made in §2); “while LayerNorm removes the dependence entirely”; “and the (a, b) fit separates the three accounts”; “Since A_t is computable online and is not flat during memorisation, it is also a progress measure in the sense of Nanda et al. (2023)”; “The kernel evolution equation above is exact, so Tiers 1–2 stand regardless”; “Runs take minutes on one GPU”; “with $a = 1$ if the finite-width correction is $O(1/N)$ ”; “so the fits are not selection-biased”; “where the published mediator does not exist”; “The dataset is $(a + b) \bmod p$ ”; “or $\lambda < 0.02$ at $\eta = 100$ ”; “computed with `torch.func`”; the pilot-figure placeholder and its caption “Baseline grokking with S_t and A_t overlaid” (the figure was dropped on 4 Sep at the group’s request). **Tier 4 moved here from the work plan:** droppable theory that bounds $\|T_t\|$ for the two-layer quadratic-activation network ($k = 3$), whose grokked solution is closed-form (Gromov, 2023), giving an analytic A_∞ and $t_{\text{grok}} \sim A_\infty/\dot{A}$; the kernel evolution equation (S5/S8) is exact, so Tiers 1–2 stand regardless. In the second 4 Sep pass (flow and set-up) the pilot figure was removed at the group’s request, “Why this regime” was merged into the end of the dispute paragraph and its clause “so the distinction Kim draws between coupled and decoupled weight decay does not arise” was dropped, and the Novelty list lost “the MLP-versus-attention comparison”. Restoring Tier 4 costs about three lines; `\linespread{0.97}` would free roughly that many if the group prefers it to cutting text.

7. **Template.** Font size, page size and the “don’t change” lines untouched; added `array`, `microtype` (character protrusion only under XeTeX), `titlesec` (tighter section spacing), tighter display skips, `\raggedbottom` and shrink-free `\topsep` (without this, a page-1 overrun of a few points shrinks the title block onto the author line instead of moving a line). Reverting lines 39–45 overflows by several lines. Submit as *G2_proposal_<SubmitterName>.pdf*.